

Week - 1 Module.

C-Programming.

① C Syntax and Basic Structure —

C is a procedural and structured programming language. Every standard C program has a precise structural layout.

Header files (`#include <stdio.h>`)

This pre-processor directive includes the standard input/output library. It allows the program to use the core functions, such as, `printf()` for output and `scanf()` for input.

Main function (`int main()`)

This is the mandatory entry point of the any C program. Execution starts at the opening brace of `main()`. Execution

ends at the ~~prog~~ terminating semicolon or return 0;

Output function (printf)

Unlike C++, C uses printf() function to display text on the screen. Double quotes let you pass format specifiers or escape sequences like \n (new line)

Semicolon (;) :- Every statement in C must end with a semicolon. It is a statement terminator for the compiler.

2. Conditions and loops:-

Conditional statements (if-else) :- used to make decisions. The if block starts when the condition is true. If it is false the program skip it or executes the optional else block.

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loops (for loops) :- To repeat a block of code a certain number of times. It is composed of initialization, condition checking and increment/decrement expressions.

Nested loops :- For star patterns to print the dimensional grid patterns we need a loop in another loop.

outer loop :- manages the rows (vertical progression)

Inner loops :- Handle the columns (print spaces or stars horizontally for the current row).

③

Programming Problems.

Solve the given star pattern Questions.

④

Square .

```
#include <stdio.h>
```

```
int main ( ) {
```

```
int n = 5;
```

```
for (int i = 1; i <= n; i++) {
```

```
int a = 1;
```

```
for (int j = 1; j <= n; j++) {
```

```
printf(" * ");
```

```
}
```

```
printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

O/p :-

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

c_programming > patternprinting > solidstar.c > main()

```
1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter no. of rows : ");
6      scanf("%d", &n);
7      for (int i = 1; i <= n; i++) // no of lines / rows -> i
8      {
9          int a = 1;
10         for (int j = 1; j <= n; j++) // no of columns -> j
11         {
12             printf("* ");
13         }
14         printf("\n");
15     }
16 }
17
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\c
?) { .\solidstar }

Enter no. of rows : 5

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka\c_programming\patternprinting> █

②

Hollow square pattern.

include <stdio.h>

int main () {

int n, m;

printf("Enter no of rows: ");

scanf("%d", &n);

printf("Enter no. of column: ");

scanf("%d", &m);

for (int i=1; i<=n; i++)

{ int a=1;

for (int j=1; j<=m; j++)

{ if (i==1 || i==n || j==1 || j==m)

printf("*");

else

printf(" ");

}

printf("\n");

}

return 0;

}

o/p —

```

* * * * *
*           *
*           *
*           *
* * * * *

```

```

1  #include <stdio.h>
2  int main()
3  {
4      int n, m;
5      printf("Enter no. of rows : ");
6      scanf("%d", &n);
7      printf("Enter no. of colume : ");
8      scanf("%d", &m);
9      for (int i = 1; i <= n; i++) // no of lines / rows -> i
10     {
11         int a = 1;
12         for (int j = 1; j <= m; j++) // no of columns -> j
13         {
14
15             if (i == 1 || i == n || j == 1 || j == m)
16                 printf("* ");
17             else
18                 printf(" ");
19         }
20     }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\c
f ($?) { .\khalisquare }

```

```

Enter no. of rows : 5
Enter no. of colume : 5

```

```

* * * * *
*       *
*       *
*       *
*       *
* * * * *

```

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka\c_programming\patternprinting> 

```


(3)

Triangle :-

include <stdio.h>

int main () {

int n=5;

for (int i=1; i<=n; i++) {

for (int j=1; j<=i; j++) {

printf (" * ");

}

printf (" \n");

}

return 0;

}

o/p -

*

* *

* * *

* * * *

* * * * *


```

1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter no. of rows : ");
6      scanf("%d", &n);
7      for (int i = 1; i <= n; i++)
8      { // no of lines / rows -> i
9          for (int j = 1; j <= i; j++)
10         { // no of columns -> j
11             printf("* ");
12         }
13         printf("\n");
14     }
15 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\c
if ($?) { .\startriangle }

```

Enter no. of rows : 5

```

*
* *
* * *
* * * *
* * * * *

```

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka\c_programming\patternprinting>

```

(14) Inverted triangle.

```
#include <stdio.h>
```

```
int main () {
```

```
int n=5;
```

```
for(int i=n; i>=1; i--) {
```

```
    for(int j=1; j<=n+1-i; j++) {
```

```
        printf(" * ");
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

O/P -

```
 * * * * *
 * * * *
 * * *
 * *
 *
```



```

1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter no. of rows : ");
6      scanf("%d", &n);
7      for (int i = 1; i <= n; i++)
8      { // no of lines / rows -> i
9          for (int j = 1; j <= n + 1 - i; j++)
10         { // no of columns -> j
11             printf("* ");
12         }
13         printf("\n");
14     }
15     return 0;
16 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\coo
leulta } ; if ($?) { .\starttriangleulta }

```

```

Enter no. of rows : 5

```

```

* * * * *

```

```

* * * *

```

```

* * *

```

```

* *

```

```

*

```

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka\c_programming\patternprinting>

```

⑤

Pyramid .

#include <stdio.h>

int main () {

int n=5;

for (int i=1; i<=n; i++) {

for (int k=1; k<=n-i; k++) {

printf (" ");

}

for (int j=1; j<=2*i-1; j++)

{

printf (" * ");

}

printf ("\n");

}

return 0;

}

O/P:-

```

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * *

```



```

1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter a number of rows : ");
6      scanf("%d", &n);
7      for (int i = 1; i <= n; i++)
8      {
9          for(int k=1;k<=n-i; k++){// for spaces
10             printf(" ");
11         }
12         for (int j = 1; j <=2*i-1; j++)
13         {
14             printf("* ");
15         }
16         printf("\n");
17     }
18     return 0;
19 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\cooder"
f ($?) { .\starpyramid }

```

```

Enter a number of rows : 5

```

```

    *
  * * *
 * * * * *
* * * * * * *
* * * * * * * * *

```

```

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka\c_programming\patternprinting> 

```

⑥

Inverted Pyramid.

#include <stdio.h>

int main () {

int n=5

for (int i = n; i >= 1; i--) {

for (int k = 1; k <= n - i; k++) {

printf (" ");

}

for (int j = 1; j <= 2 * i - 1; j++)

{

printf ("*");

}

printf ("\n");

}

return 0;

}

O/P:-

```

*****
 ***
  **
   *
    *

```



```

1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter number of rows: ");
6      scanf("%d", &n);
7      for (int i = n; i >= 1; i--)
8      {
9          // Spaces
10         for (int k=1;k<=n-i;k++)
11         {
12             printf(" ");
13         }
14         // Stars
15         for (int j=1;j<=2*i-1;j++)
16         {
17             printf("* ");
18         }
19         printf("\n");
20     }
21     return 0;
22 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\hg113\OneDrive\Desktop\cooder kaka> cd "c:\Users\hg113\OneDrive\Desktop\code
tedpyramid } ; if (\$?) { .\starinvertedpyramid }

Enter number of rows: 5

```

* * * * *
 * * * * *
  * * * *
   * * *
    *

```

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⑦

Half Diamond

#include <stdio.h>

int main() {

int n=5

// Upper Half

for (int i=1; i<=n; i++)

{ for (int j=1; j<=i; j++) {

printf("* "); }

printf("\n");

}

// Lower Half

for (int i=n-1; i>=1; i--)

{ for (int j=1; j<=i; j++) {

printf("* ");

}

printf("\n");

}

return 0;

}

o/p: - *

```

* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```



```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6
7      printf("Enter number of rows: ");
8      scanf("%d", &n);
9
10     // Upper Half
11     for (int i = 1; i <= n; i++)
12     {
13         for (int j = 1; j <= i; j++)
14         {
15             printf("* ");
16         }
17         printf("\n");
18     }
19
20     // Lower Half
21     for (int i = n - 1; i >= 1; i--)
22     {
23         for (int j = 1; j <= i; j++)
24         {
25             printf("* ");
26         }
27         printf("\n");
28     }
29
30     return 0;
31 }
```

```
PS C:\Users\hg113\OneDrive\Desktop\cooder_kaka\c_programming\patternprinting> cd "C:\Users\hg113\OneDrive\Desktop\cooder_kaka\c_programming\patternprinting" & gcc Diamond.c -o halfDiamond } ; if ($?) { .\halfDiamond }  
Enter number of rows: 5
```

```
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * *  
* * * *  
* * *  
* *  
*  
*
```

```
PS C:\Users\hg113\OneDrive\Desktop\cooder_kaka\c_programming\patternprinting> 
```


⑧

Diamond:.

include <stdio.h>

int main () {

int n=5;

// Upper Pyramid

for (int i=1; i<=n; i++) {

// Spaces

for (int j=1; j<=n-i; j++) {

printf(" "); }

// Stars

for (int k=1; k<=2*i-1; k++) {

printf("*"); }

printf("\n");

}

// Lower Inverted Pyramid

for (int i=n-1; i>=1; i--) {

// Spaces

for (int j=1; j<=n-i; j++) {

printf(" "); }

// Stars

for (int k=1; k<=2*i-1; k++) {

printf("*"); }

printf("\n");

}

return 0;

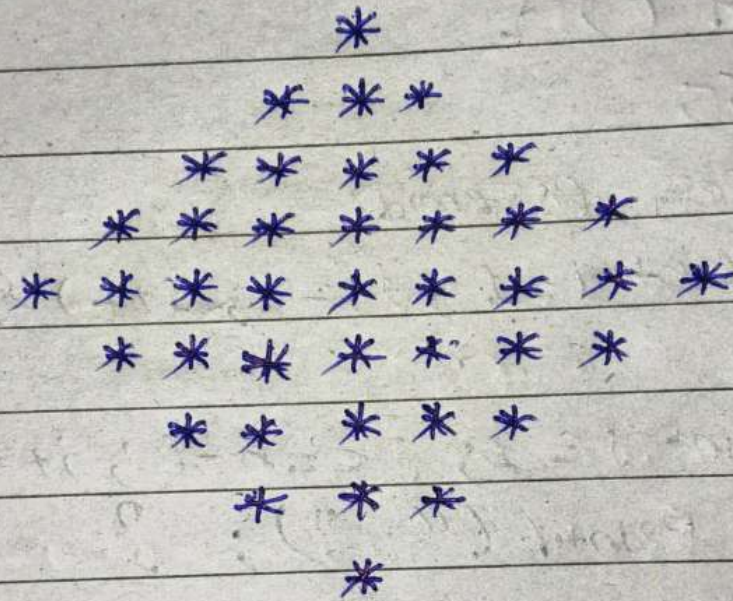
}

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```
1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      printf("Enter number of rows: ");
6      scanf("%d", &n);
7      // Upper Pyramid
8      for (int i = 1; i <= n; i++)
9      {
10         // Spaces
11         for (int j = 1; j <= n - i; j++)
12         {
13             printf(" ");
14         }
15         // Stars
16         for (int k = 1; k <= 2 * i - 1; k++)
17         {
18             printf("* ");
19         }
20         printf("\n");
21     }
22     // Lower Inverted Pyramid
23     for (int i = n - 1; i >= 1; i--)
24     {
25         // Spaces
26         for (int j = 1; j <= n - i; j++)
27         {
28             printf(" ");
29         }
30         // Stars
31         for (int k = 1; k <= 2 * i - 1; k++)
32         {
33             printf("* ");
34         }
35         printf("\n");
36     }
37     return 0;
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\hg113\OneDrive\Desktop\cooder\kaka\c_programming\patternprinting> cd "c:\Users\hg113\OneDrive\Desktop\cooder\kaka\c_programming\patternprinting" & gcc diamond.c -o fulldiamond } ; if ($?) { .\fulldiamond }
```

Enter number of rows: 5

```

      *
    * * *
  * * * * *
* * * * * *
* * * * * * *
  * * * * *
    * * * *
      * * *
        *
```

```
PS C:\Users\hg113\OneDrive\Desktop\cooder\kaka\c_programming\patternprinting> 
```